Does inferred demonstrator expertise transfer to a new environment? Learning from demonstrations in multi-armed bandits

Julio Martinez Michael C. Frank Nick Haber

October 26, 2026

Project links

- **Live experiment:** https://asteroid-mining-game-31128277324.us-east4.run.app/ (the version deployed for this study; take the study exactly as participants do)
- **Experiment code:** https://github.com/martinezjulio/AsteroidMiningAdenture, tag `[TODO: 6.1]`
- **Modelling and analysis code:** https://github.com/martinezjulio/SocialTransformer, tag `[TODO: 6.1]`
- **Transformer checkpoint:** sha256 `[TODO: 6.1]`, uploaded with the registration
- **Preregistration:** OSF registration *(registered `[TODO: 6.3]`, before recruitment)*

AI use statement

I used Claude Code (various version of Sonnet, Opus 5.5 and Fable 5.1) to help me implement the experiment, code, the modeling and analysis pipeline and tests. Claude Code was also used to set up this document, its Quarto structure, rendering and the tables of pipeline numbers. The text of this registration however is my own.

1 Introduction

This study asks whether a social learner's theory of mind representations of a social partner, i.e their latent environmental beliefs and experience, shape the learner's choices, and results in improved outcomes in a new environment with prior exposure to the same social partner. We study this in a multi-armed ban-

dit task where participants both play and watch demonstrations of simualated partners with different levels of experience.

2 Task and design

2.1 Materials

The task. The task is a 12-armed bernoulli bandit themed as a asteroid mining adventure game. Participants mine by shooting asteroids which represent the arms of the game. Asteroids represent the arms of the game and shooting them result in reward of 0 or 1 based on a hidden underlying payoff probability. All bandits re-used the same set of payoff probabilities from $\{0, 0.1, 0.2, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.75, 0.8\}$ where different bandits are permutations of these payoffs.

Games. Conditions in the task are made up of games. Each game is a either a watch play game or a watch game. Play games allow exactly 6 trials of play on the bandit done by the participant. Watch games also alloow exactly 6 trials but they are done by a simulated partner which the participant watches. Both types of games show the outcomes of the trials both the choice that was made and the reward outcome received. In addition there is a choice game that only applies to the social conditions. The choice game is a single trial where participants are asked to select one of the two demonstrator partners they watched during the watch games.

Partners. There are two types of demonstrator partners that can be watched during the watch games: an expert or a novice. Both are Thompson samplers where the novice has no prior experience and hence starts from uniform prior beliefs whereas the expert has 128 trials of prior experience that forms their beleifs over the bandit. Experience however is hidden to the participant and can only be inferred from oberving their demonstrations and outcomes. One key detail participants are informed about is that partners that have more experience have more experience across all bandits and vice versa for novice partners. Thompson samplers for the demonstrators are generated prior to each watch game and only filters for the expert's choices being better in expectation than the novice's on that bandit however this doesn't prevent reward outcomes from favouring the novice.

2.2 Procedure

Order of events.

1. Consent
2. Instructions
3. Practice session
4. 3 Main conditions
5. Return screen

The practice session is a sequence of games defined above as [Play, Watch, Play] all on the same bandit for familiarization and not analyzed.

Conditions. There are 3 conditions. 2 are social conditions made up of watch and play games. The third is a non-social or solo condition made up of entirely play games. Every participant sees every condition in random order. Each condition is split into rounds. Each round repeats the same set of games again but in a new bandit with new payoffs. In the social conditions both an expert and a novice demonstrator are watched in each round. The order of the type of demonstrator watched first is counterbalanced across participants. The social conditions end with a choice game whereas the solo condition does not. The table below shows the structure of each condition.

Condition	Each round	Choice game
expert-first	watch expert, **play**, watch novice, **play**	true
novice-first	watch novice, **play**, watch expert, **play**	true
solo	**play**, **play**, **play**, **play**	false

Outcomes. Three outcomes are measured in the play games: the mean of the true payoffs of the chosen arms, the mean of observed rewards, and the mean entropy of the arm choices (in bits).

Randomization. Condition order, payoff probabilities in each bandit (for each round), outcome rewards themselves, bandit background images in the game (corresponding to each bandit), asteroid images for each arm for each bandit, and the order partners are presented in choice games are all randomized.

3 Hypotheses

The list below shows each confirmatory claim, both primary and secondary. For each confirmatory claim the estimand, the model from which it is derived and the accompanying rule are listed. The primary claims use the H labels (i.e. H1 through H5) and the secondary claims use the S label (i.e. S1 through S4). The contrast expressions for the H1, H2, H3 and H5 claims are included; the models from which these contrasts are derived are specified in Section 4. In the contrast expressions, `emm <- emmeans(fit, ~ condition * play * round)` is a twelve-cell grid of the fitted descriptive model, `m(sel)` is the weight vector that averages the selected cells, and `demo` is the demonstrator watched before the play game (`"none"` in the solo condition), as in `src/r/descriptive.R`.

Primary Claims:

- **H1.** Participants play better right after watching the expert than right after watching the novice.

  - Estimand: mean payoff and mean entropy of the play games that follow an expert demonstration minus those that follow a novice demonstration, each averaged over both rounds and both social conditions (four cells against four).
  - Model: the mixed-effects model of play outcomes, Section 4.1.
  - Rule: Satterthwaite t-test on the contrast, two-sided $\alpha = 0.05$ with 95% CI; predicted sign payoff $+$, entropy $-$; Holm over the two outcomes.
  - Contrast: `contrast(emm, list(w = m(demo == "expert") - m(demo == "novice")))`
- **H2.** Participants play better after watching the novice demonstrator the second time, in round 2, than after watching the novice the first time, in round 1.
  - Estimand: mean payoff and mean entropy of the after-novice play games in round 2 minus the same games in round 1, averaged over the two social conditions.
  - Model: the mixed-effects model of play outcomes, Section 4.1.
  - Rule: Satterthwaite t-test on the contrast, two-sided $\alpha = 0.05$ with 95% CI; predicted sign payoff $+$, entropy $-$; Holm over the two outcomes.
  - Contrast: `contrast(emm, list(w = m(demo == "novice" & round == 2) - m(demo == "novice" & round == 1)))`
- **H3.** In round 2 participants concentrate their choices on fewer arms than in round 1.
  - Estimand: mean entropy of the six choices of a play game in round 2 minus round 1, averaged over all play games of both social conditions.
  - Model: the mixed-effects model of play outcomes, Section 4.1.
  - Rule: Satterthwaite t-test on the contrast, two-sided $\alpha = 0.05$ with 95% CI; predicted sign entropy $-$; one outcome, no correction.
  - Contrast: `contrast(emm, list(w = m(demo != "none" & round == 2) - m(demo != "none" & round == 1)))`
- **H4.** After watching the expert and novice demonstrators each in both rounds 1 and 2, participants choose the expert more often than the novice.
  - Estimand: the probability that a participant chooses the expert in the choice game that ends a social session, estimated with a participant-level intercept; the claim is that it exceeds one half.
  - Model: the mixed-effects model of partner choice, Section 4.2.
  - Rule: Wald z-test on β_0, two-sided $\alpha = 0.05$ with 95% CI; predicted sign $\beta_0 > 0$ on the logit scale.
  - Fit: `glmer(chose_expert ~ 1 + (1 | participant), family = binomial)`
- **H5.** Participants play better after watching the expert, in both rounds, than solo participants do in their final game.
  - Estimand: mean payoff of the play games that follow an expert demonstration minus mean payoff of the solo condition's fourth and final play game of the round, averaged over rounds. The solo game

comes after 18 own choices; the after-expert game after 6 (expert-first) or 18 (novice-first) revealed choices, so solo has at least as much information in every comparison.
  - Model: the mixed-effects model of play outcomes, Section 4.1.
  - Rule: Satterthwaite t-test on the contrast, two-sided $\alpha = 0.05$ with 95% CI; predicted sign payoff +; one outcome, no correction.
  - Contrast: `contrast(emm, list(w = m(demo == "expert") - m(demo == "none" & play == "second")))`

Secondary Claims:

- **S1.** Arm choices are predicted better by models that use the estimate of the expert's beliefs than by models that use the estimate of the most recently watched demonstrator's beliefs (S1a) or of the novice's beliefs (S1b).
  - Estimand: out-of-sample predictive accuracy, ΔELPD, of the full policy model reading the expert's belief minus the same model reading the last-watched demonstrator's belief (S1a) and minus the same model reading the novice's belief (S1b), with the standard error paired over participants.
  - Models: HOS of expert vs recent (S1a) and vs novice (S1b).
  - Rule: not a test; each part ΔELPD, Equation 7, > 2 paired SE, reported with its SE; both required.
- **S2.** Arm choices are predicted better by models that use the estimate of the expert's beliefs than by models that simply count the successes and failures of 1. the most recent demonstrator they watched (S2a), 2. the expert demonstrator they watched (S2b), or 3. every demonstrator they watched (S2c).
  - Estimand: ΔELPD of the full model minus the same model with the expert's belief replaced by a Beta count of the visible choices of the last demonstration (S2a), of the expert's demonstration on this bandit (S2b), and of every choice seen on this bandit this round (S2c).
  - Models: HOS of expert vs naive-recent (S2a), vs naive-expert (S2b) and vs naive (S2c).
  - Rule: not a test; each part ΔELPD > 2 paired SE, reported with its SE; all required.
- **S3.** Arm choices are predicted better by models that use the estimate of the expert's beliefs alone than by models that use only own experience (S3a). Similarly, models that use the estimate of the expert's beliefs as well as own experience predict choices better than models that use only own experience (S3b).
  - Estimand: ΔELPD of the model with the expert's belief and no own outcomes minus the model with own outcomes and no belief (S3a), and of the full model minus the model with own outcomes and no belief (S3b).
  - Models: HS vs HO (S3a), HOS vs HO (S3b).

  - Rule: not a test; each ΔELPD > 2 paired SE, reported with its SE; both required.
- **S4.** Participants are more influenced by the expert's belief in round 2 after watching the novice than in round 1 after watching the novice.
  - Estimand: in the switch model, the group-level change from round 1 to round 2 in the weight placed on the expert's belief on play games that follow the novice, on the logit scale, $\Delta = \delta_N$.
  - Model: Switch-by-round, Eq. 6.
  - Rule: posterior interval, not a test; 95% ETI of Δ excludes 0 on the predicted side.

Rules and verdicts. The Holm correction applies across the outcomes within a hypothesis. Verdicts are *supported*, *not supported* or *contradicted*. No smallest effect of interest is registered.

4 Models and analysis

4.1 Mixed-effects model of play outcomes

Model. One linear mixed model per outcome (payoff, reward, entropy), fitted with `lmerTest::lmer` on one row per play game (`src/r/descriptive.R`):

```
outcome ~ condition * play * round + (1 | participant)
```

- `condition`: factor with three levels: expert-first, novice-first and solo; group-level.
- `play`: factor with two levels, referring to the 1st and 2nd play games in the social conditions, and 2nd and 4th play games of the solo condition; group-level.
- `round`: factor with two levels, 1 and 2, each on a new bandit; group-level.
- `condition * play * round`: the three main effects and all their interactions, under sum-to-zero coding; all group-level.
- `(1 | participant)`: a random intercept per participant.

In the solo condition the 2nd and 4th play games are used to match the social condition's 1st and 2nd play games. This choice was made in order to have the same number of trials of observations before these play games in both the social and solo conditions.

Contrasts. The model results in 12 cell means per outcome type (payoff, reward, entropy), one per combination of condition, play and round. Each cell is read off the fit with `emm <- emmeans(fit, ~ condition * play * round)` with Satterthwaite degrees of freedom. Each hypothesis is the mean of one set of cells minus the mean of another, computed as `contrast(emm, list(w = m(A) - m(B)))` with `A` and `B` for each claim given in Section 3.

Reward. In addition to payoff and entropy we also report observed reward with its 95% confidence interval, for the same contrasts of payoff (H1, H2 and

H5). It reflects the same quantity but noisier.

4.2 Mixed-effects model of partner choice

Model. A single logistic mixed model fitted with `lme4::glmer` on one row per social condition per participant (the final choice game in each, so 2 rows total per participant), with `chose_expert` equal to 1 if the participant chose the expert (`src/r/partner_choice.R`):

```
chose_expert ~ 1 + (1 | participant), family = binomial
```

- `1`: the intercept, the group-level log-odds that the expert is chosen.
- `(1 | participant)`: a random intercept per participant, a normal deviation of each participant's own log-odds; the only per-participant term.
- `family = binomial`: the outcome is 0/1, modelled as Bernoulli with its probability on a log-odds scale.

Reporting. The intercept and its 95% confidence interval are reported on the logit scale z and the inverse logit (probability scale). In addition the mean raw rate of expert picks is reported with a 95% Wilson score interval, for all sessions and for each condition.

4.3 Policy models

Setup. A policy model is a generative model of a participant's choices during the play games. The first part of the pipeline uses a transformer to generate predictions of each demonstrator's beleifs over the payoffs of the arms of the bandit, given each of the demonstrators observed choices and outcomes. These are called the readouts. The second part of the pipeline determines the different terms that enter the policy model, including the demonstrators' beleifs, the participant's own outcomes, and the participant's own previous step's choice and reward outcome. Each of these terms are combined into what becomes the policy model, and is fitted to the participant's choices and outputs a log-likelihood of the choices given the model computed as a soft-max.

The policy models are fitted to every shot of the play games of the two social conditions (expert-first and novice-first). For each of the two conditions there two rounds, two play games, six shots, making 48 shots total per participant. Each shot is a trial t of participant i choosing arm a_t from $K = 12$ arms. The readouts from the transformer are the parameters of a Beta distribution per arm k, $(\alpha_k^{\text{demo}}, \beta_k^{\text{demo}})$. The participant's own outcomes are the counts of successes and failures on each arm k in the current play game, s_k^{curr} and f_k^{curr}, and in the earlier play game of the same round, s_k^{prev} and f_k^{prev}. The previous arm chosen is a_{t-1} with its reward outcome $r_{t-1} \in \{0, 1\}$.

The full model, HOS. The full model includes all terms: the perseveration or habit term (H), the own-experience term (O) and the social-belief term built on the readouts (S), hence the name HOS. Each term carries its own weight.

$$\alpha_k = \kappa_i\, \alpha_k^{\text{demo}} + \lambda_{\text{w}}\, s_k^{\text{curr}} + \lambda_{\text{c}}\, s_k^{\text{prev}}, \qquad \beta_k = \kappa_i\, \beta_k^{\text{demo}} + \lambda_{\text{w}}\, f_k^{\text{curr}} + \lambda_{\text{c}}\, f_k^{\text{prev}}, \tag{1}$$

Each arm's belief distrubiton is a Beta(α_k, β_k) distributiopn. Under Thompson sampling an arm is chosen with the probability that its draw from this distribution is the highest among the twelve arms. This is

$$p_k = \int_0^1 f_k(x) \prod_{j \neq k} F_j(x)\, dx, \tag{2}$$

with f_k and F_k the probability density function (pdf) and cumulative distribution function (CDF) of Beta(α_k, β_k), respectively. The log of p_k is the logit of arm k. A win-stay lose-shift tilt adds a bonus, with sign depending on the last reward, to the logit of the previous arm,

$$\ell_k = \log p_k + (\rho_{0,i} + \rho_{\text{win},i}\, r_{t-1})\, \mathbb{1}[k = a_{t-1}], \tag{3}$$

The final choice is a softmax over the logits, at unit temperature, with a fixed lapse,

$$\pi_k = \frac{\exp \ell_k}{\sum_j \exp \ell_j}, \qquad P(a_t = k) = (1 - \varepsilon)\, \pi_k + \frac{\varepsilon}{K}, \qquad \varepsilon = 0.02. \tag{4}$$

Without the readout, the readout term of Equation 1 is replaced by 1, so $\alpha_k = 1 + \lambda_{\text{w}}\, s_k^{\text{curr}} + \lambda_{\text{c}}\, s_k^{\text{prev}}$ and likewise β_k with the failure counts, or $\alpha_k = \beta_k = 1$ for perseveration alone. Without the win-stay lose-shift $\ell_k = \log p_k$ and $\pi_k = p_k$.

How parameters are shared across participants and conditions is described in the table below.

- κ_i, $\rho_{0,i}$, $\rho_{\text{win},i}$: hierarchical, one value per participant drawn from a group mean, κ on a log scale.
- λ_{w}, λ_{c}: group-level.
- ε: fixed at 0.02.

Every prior is listed below.

The ladder. Every other model is Equation 1 to Equation 4 with a term replaced or removed. HOS isthe model with all terms. Models missing the habit term drop the H, and similarly for O and S. The models of S1 to S3, each described as a change from HOS:

Model	Terms included
HOS	readout of the expert (S), own counts (O), habit tilt (H)
recent	same as HOS but with the readout of the last-watched demonstrator
novice	same as HOS but with the readout of the novice
naive-recent	same as HOS but with Beta$(1,1)+\kappa_i\times$ the visible shots of the last demonstration in place of the readout
naive-expert	same as HOS but with Beta$(1,1)+\kappa_i\times$ the visible shots of the expert's demonstration on current bandit in place of the readout
naive	same as HOS but with Beta$(1,1)+\kappa_i\times$ every shot seen on the bandit in current round including play and watch games
HS	same as HOS without the own counts
HO	same as HOS without the readout

Random is uniform choice with nothing fitted and included as a floor.

The Switch models. Two HOS policies sharing the same parameters for the perseveration, own-experience and social terms (ρ, λ and κ_i), one using the expert's readout and the other the novice's, are mixed,

$$P(a_t = k) = (1-\varepsilon)\left[\omega_t\,\pi_k^E + (1-\omega_t)\,\pi_k^N\right] + \frac{\varepsilon}{K}, \tag{5}$$

where ω_t is the participant's weight on the expert's policy for that play game, $\omega_{E,i}$ when the game follows the expert and $\omega_{N,i}$ when it follows the novice. Each is a participant-level weight and is hierarchical on the logit scale (e.g. $\operatorname{logit}\omega_{E,i} = \mu_E + \tau_E z_{E,i}$). Switch-by-round lets each weight change based on the round at the group level,

$$\begin{aligned} \operatorname{logit}\omega_{N,i,r} &= \mu_N + \tau_N z_{N,i} + \delta_N x_r, \qquad x_1 = -\tfrac{1}{2},\ x_2 = +\tfrac{1}{2}, \\ \Delta = \delta_N &= \operatorname{logit}\omega_N(2) - \operatorname{logit}\omega_N(1), \end{aligned} \tag{6}$$

and likewise for E with δ_E.

Model	Terms included
Switch	two HOS policies, one on the expert's readout and one on the novice's, mixed by Equation 5
Switch-by-round	same as Switch with the round term Equation 6

Priors. Hierarchical parameters are non-centred. The group mean is μ, the group spread is τ and the standard-normal participant deviation is z_i. They

are combined as $\mu + \tau z_i$ on the scale named in the table below. Parameters with the participant subscript i are hierarchical. The rest are group-level. The exploratory variants of Section 5 add parameters whose priors are in the fitting code at the registered tag.

Parameter	Scale	Prior
κ_i	log	$\mu_\kappa \sim \mathcal{N}(0, 0.5)$, $\tau_\kappa \sim$ HalfNormal(0.3)
λ_{w}	log	$\mathcal{N}(0, 1.5)$, group level
λ_{c}	linear	HalfNormal(4), group level; fixed at 0 in HOS-reset
$\rho_{0,i}$	linear	$\mu \sim \mathcal{N}(0, 2)$, $\tau \sim$ HalfNormal(1)
$\rho_{\mathrm{win},i}$	linear	$\mu \sim \mathcal{N}(0, 2)$, $\tau \sim$ HalfNormal(1)
ε	linear	fixed at 0.02
$\omega_{E,i}$, $\omega_{N,i}$	logit	$\mu \sim \mathcal{N}(0, 1.5)$, $\tau \sim$ HalfNormal(1)
δ_E, δ_N	logit	$\mathcal{N}(0, 1)$, group level

4.4 Fitting and model comparison

Thompson probabilities. Equation 2 has no closed form. It is evaluated by trapezoidal quadrature on a uniform grid of the logit scale, over a dense grid of κ, λ_{w} and λ_{c} (accurate to about 10^{-7}) and interpolated, so every model shares one likelihood.

Sampling and gate. Every model is fitted with NUTS in NumPyro, four chains of 1,000 warm-up and 1,000 sampling iterations at target acceptance 0.9. A fit passes when $\hat{R} < 1.01$ and bulk and tail effective sample sizes are at least 400 on every parameter, with no divergent transitions. A failing fit is refitted at 2,000 + 2,000 iterations at acceptance 0.95, then 4,000 + 4,000 at 0.99; a fit that fails the last rung is not reported.

Comparison. Models are compared by expected log predictive density estimated with Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO) from the pointwise log-likelihoods of the posterior draws, summed over shots. For models A and B, with d_i the sum over participant i's shots of the pointwise difference and N the number of participants,

$$\Delta\mathrm{ELPD}_{AB} = \sum_{i=1}^{N} d_i, \qquad \mathrm{SE}_{AB} = \sqrt{N}\,\mathrm{sd}(d_1, \ldots, d_N), \tag{7}$$

so the standard error is paired over participants. A comparison supports A when $\Delta\mathrm{ELPD}_{AB} > 2\,\mathrm{SE}_{AB}$. A shot whose Pareto k exceeds 0.7 has an unreliable importance estimate; for HOS and its five rivals of S1 and S2, every session holding such a shot is refitted without that session and its shots enter by their exact held-out log predictive density.

Verdicts. A multi-part claim is *supported* only if every part clears the rule, *contradicted* if any part favours the rival by more than two standard errors, and *not supported* otherwise; every part's verdict is reported. A claim is *not evaluable* when a fit it needs failed the gate or its recovery check below failed; the comparison is still reported.

Recovery checks. Two simulation checks run on the frozen code before any verdict is read. (a) Parameter recovery of δ_N: 25 datasets of 100 simulated participants are generated from posterior draws of the Switch-by-round fit to the analysed data, each played through the full experiment with the transformer re-read before every play game, and refitted with the registered pipeline; the check passes when the 95% intervals cover the true δ_N and exclude zero on its side in at least 22 of 25. (b) Model recovery for S1a (HOS vs recent) and S2a (HOS vs naive-recent): five datasets are generated from each model of the pair and both are fitted to each; the check passes when the generating model wins a majority of its own datasets and at most a minority of its rival's, an undecided dataset counting for the rival. The S1a and S2a checks gate S1 and S2; the δ_N check gates S4.

4.5 Posterior predictive checks

Free-running check. The fitted HOS and Switch-by-round models are each made to play the experiment and their behaviour compared with the participants'. At each of 50 posterior draws spaced evenly over the chains, every participant replays their own two social sessions with their drawn parameters: the fitted policy takes every shot of every play game, the demonstrations are the ones the participant saw, and the transformer is re-read before each play game so that the readout entering a round's second play game reflects the simulated first one. Five statistics are computed on each replay exactly as on the human data, each a mean over participants of a per-participant mean:

1. the after-novice payoff rise from round 1 to round 2 (H2's quantity);
2. the entropy fall from round 1 to round 2 over the social play games (H3's quantity);
3. the novice-first zig-zag, payoff after the expert minus after the novice within novice-first sessions;
4. the share of shots that repeat the previous arm after a win;
5. the same share after a loss.

For each statistic the 2.5th and 97.5th percentiles over the 50 replays form a 95% band, and the check passes when the human value lies inside it. A missed target does not change any verdict: the checks are registered as a description of what the winning model can and cannot reproduce, reported for every target and both models in a figure with the human value against the band, and discussed as such.

Split-half generative check (exploratory). The same two models are fitted on a random half of the participants (fixed seed) and the other half is replayed

at each of the 50 draws with parameters drawn fresh from the fitted group-level hierarchy rather than their own posteriors, so the band shows what the population model predicts for participants it has never seen. The five statistics of the held-out half are compared with the band in the same way and reported beside the main check.

5 Exploratory analyses

The analyses below are ones that may be run, not a commitment to run every one of them. They are written down as a closed list so that whichever of them appear in the paper were planned before the data arrived rather than invented afterwards; an analysis in the list that is not run is simply not reported. None of them carries a decision rule: each is reported with its estimate and 95% interval, and where an expectation is stated it is a prior belief that frames the reading, not a test. Anything not in Section 3 or in this list is post hoc; if it appears in the paper it is labelled post hoc and stated to have been chosen after seeing the data.

	Analysis	What is reported	Expectation
E1	after-expert round change	round 2 minus round 1 in the after-expert cells, from the model of Section 4.1, all outcomes	no fall; may rise
E2	the other social cells vs solo's final game	H5's comparison with the after-novice cells (E2a) and with the social cells averaged over demonstrators (E2b) in place of the after-expert cells, and H5 within round 2 alone (E2c) (payoff)	after-novice near zero; the average positive and smaller than H5; round 2 like H5
E3	the after-novice improvement vs solo practice	H2's difference minus the same round difference in solo's first and second play, from the model of Section 4.1, all outcomes; the solo term alone is reported as E3s	payoff positive; entropy ambiguous, since solo entropy also falls with practice
E4	reward	the reward version of every contrast above, H1, H2, H5 and E1 to E3, with its 95% interval (Section 4.1)	same sign as payoff, wider intervals

	Analysis	What is reported	Expectation
E5	condition breakdown of the after-novice round change	H2's contrast within expert-first and within novice-first separately; in novice-first the round 2 after-novice game is on a new bandit before the expert has been seen on it, so any improvement there is carried over from round 1	none stated
E6	sensitivity of H1, H2, H3 and H5	the model of Section 4.1 refitted two ways, each reporting estimate, interval and uncorrected p-value without Holm or verdict: (a) random slopes for round and play by participant, coded $-\frac{1}{2}$ and $+\frac{1}{2}$, reduced to uncorrelated slopes and then to the intercept-only model if singular or non-convergent, the specification reached reported; (b) session order as a numeric covariate in the fixed part, cell means taken at its mean	same verdicts
E7	reported ladder rungs	paired ΔELPD with its SE, Equation 7, as rows of the choice stage: the ladder's difference from HOS for OS (habit), HOS+persev (perseveration) and HOS-reset (own outcomes not carried between play games), and the two paired rows HOS minus HS (own experience on top of the readout) and S minus O (social belief against own experience)	habit positive; S over O positive; the rest near zero

	Analysis	What is reported	Expectation
E8	δ_E	the round change of the weight on the expert's belief after the expert, Equation 6, as its posterior mean and 95% ETI from the Switch-by-round fit	no fall
E9	Switch-by-round vs Switch	ΔELPD of the round term; beside it, model recovery for this pair run as in check (b) of Section 4.4	positive
E10	robustness of S4	three refits, each reporting posterior means with 95% ETIs: Switch-by-round with a group-level round trend on $\log \kappa$ (Switch-by-round+ktrend: δ_κ and δ_N); HOS with the lapse rate free at group level (HOS+eps: ε and the group-level κ); Switch-by-round with the lapse rate free (Switch-by-round+eps: ε and δ_N)	Δ unchanged in sign and interval
E11	the unseen-expert readout	the five analyses (a) to (e) of the table below	the expert's readout matters before the expert is seen on the bandit, in round 2
E12	recency in the partner choice	the model of Section 4.2 with condition added, `chose_expert ~ condition + (1 | participant)` under sum-to-zero coding, so the condition term is half the expert-first minus novice-first difference on the logit scale; in expert-first the novice was watched last, in novice-first the expert, so a term near zero means the choice tracks expertise rather than recency	near zero

Contrasts of E1 to E3. Written as in the table of Section 3; `demo == "none"` selects the solo cells and `play == "second"` solo's final game of a round.

	Contrast
E1	`m(demo == "expert" & round == 2) - m(demo == "expert" & round == 1)`
E2a	`m(demo == "novice") - m(demo == "none" & play == "second")`
E2b	`m(demo != "none") - m(demo == "none" & play == "second")`
E2c	`m(demo == "expert" & round == 2) - m(demo == "none" & play == "second" & round == 2)`
E3	`(m(demo == "novice" & round == 2) - m(demo == "novice" & round == 1)) - (m(demo == "none" & round == 2) - m(demo == "none" & round == 1))`
E3s	`m(demo == "none" & round == 2) - m(demo == "none" & round == 1)`

E11, the unseen-expert readout. In the novice-first condition, the first play game of round 2 follows the novice's demonstration on a new bandit, before the expert has shot on it; the readout of the expert's belief there can draw only on the expert's earlier bandit and on the novice's shots. Whether that readout is still the better account of choices, and what it contains, is E11. The five analyses are fixed outputs of the pipeline (`src/analysis/unseen.py` and the Switch-8 fit), carry no decision rule, and are read with the caveat registered beside S4 (Section 4.3). The expectation is that the expert's readout matters before the expert is seen, in round 2 but not in round 1.

	Analysis	What is computed
a	what the readout contains	per readout (expert, novice), bandit state (expert already seen on it or not) and round, the readout entering a play game is scored four ways: its confidence (the total Beta count), the Spearman rank correlation of its arm means with the true payoffs, the true payoff of its favourite arm, and the entropy of the Thompson policy on it at $\kappa = 1$; each a mean over sessions with its standard error
b	the readout against copying	on after-novice play games, by expert-seen and round: the rank correlation across arms between the unseen-expert readout and the copying belief, $\mathrm{Beta}(1 + \text{wins}, 1 + \text{losses})$ from the novice's visible shots, and of each with the truth

	Analysis	What is computed
c	the ELPD split	two paired comparisons from the fits of Section 4.4, cut into the after-novice cells of expert-seen × round with Equation 7 per cell: HOS against naive-recent (S2a's pair) and HOS against recent, which on these shots reads the novice's belief (S1a's pair, so the cells add up to S1a)
d	the exploit statistic	model-free, on after-novice shots by expert-seen and round: the share of a participant's shots on arms where the novice's demonstration succeeded at least once, beside the share of such arms among the twelve (chance) and the share of the participant's shots on arms the novice tried at all; mean and standard error over participants. Exploiting the evidence raises the first share above chance without raising the third
e	the eight-weight switch model	Switch-8 (Section 4.3) gives one group-level weight on the expert's policy per last demonstrator × round × condition, laid out in session order so that the after-novice weight in round 2 of novice-first, where the expert is unseen, can be read against the same cell in round 1 and against round 2 of expert-first, where the expert has been seen

6 Sampling plan

6.1 Existing data

Registration prior to creation of the confirmatory data. Several pilots of this task were run to work out the arm payoffs, the expert demonstrators' experience level, the framing and reward labelling, and the labelling of demonstrators, zones and expeditions throughout the game. The last pilot (pilot E, 38 participants, three sessions each) ran the final version of the task, and the models and hypotheses in this document were developed on it. Pilot E numbers appear below only as descriptive context and as the assumptions of the sample-size analysis, and are labelled as such; none of the pilot data enters the confirmatory dataset, and no pilot participant may take part in the full study. Pilot participants were not asked for consent to data sharing, so the pilot trial data are not released; the IRB protocol permits their internal use for the sample-size analysis, whose inputs, method and outputs are reported in Section 6.3 and whose output tables are attached to the registration. A reader can check the sample-size chain from those tables but cannot re-run the simulation. No further pilot precedes the full run; if the task code changes before recruitment (for example new log fields), the new fields are checked by the experimenter, not piloted.

6.2 Participants and sampling

Population and recruitment. Participants are recruited on Prolific with these prescreens: resident in the United States, aged 18 to 50, fluent in English, and no colour-vision deficiency. The colour-vision prescreen is Prolific's stored profile self-report, unverified and possibly old, so it removes people who know they are colour blind and no more; the two partner starships and the asteroids are told apart partly by colour but also by name and image, and the game asks no colour-vision question of its own (the stored answer is exclusion rule X2 of Section 6.4). Two platform filters are added: an approval rate of 95% or higher on previous studies and at least 10 previous submissions. Anyone who took part in any pilot of this task or in the related experiments of our earlier work is excluded in advance through a Prolific participant group of their IDs. The target sample size N is set in Section 6.3.

Stopping rule. Recruitment runs in batches on Prolific until $N = 350$ participants (Section 6.3) have all three sessions complete and survive every exclusion rule of Section 6.4. After each batch the exclusion script is run to count survivors, and that count is the only thing looked at before the end: no confirmatory outcome, contrast or model is computed on any part of the data until recruitment has closed. Batches are sized so that the last one is expected to reach N with a small overshoot; every participant in the final batch who passes the rules is analysed, so the analysed sample may exceed N by a few and is never trimmed to hit it exactly. Recruitment is not stopped early for any reason other than reaching N, is not bounded by a spending limit, and is not extended beyond N to chase a result. Bonuses are paid after each batch.

Payment. A base payment of $2, which assumes completion in 10 minutes at $12 per hour, plus the performance bonus of 2 cents per point earned in play games, accumulated across sessions and shown in the game as a running total (watch games and the choice game earn nothing), paid in bulk after each batch. In pilot E the median time from start to finish was 12 minutes (quartiles 10 to 15). Participants who decline consent on the first page return their submission unpaid and are not counted toward the sample.

Consent and data sharing. Participants consent on the first page. The one consent covers both taking part and the de-identified public sharing of the data; there is no separate sharing item, so every analysed participant's anonymised trial data are released (Section 7). Age and sex, taken from Prolific's profile export, are summarised by the preprocessing script over the analysed sample (mean, SD and range of age; counts by sex) and reported descriptively; they enter no analysis and are not released per participant. The consent text is in the experiment repository at the registered tag.

6.3 Power analysis and sample size

Method. Power for H1 to H5 is estimated by simulation on pilot E, whose 38 participants are the only source of effect sizes and noise for this task, so every

number below assumes the full run resembles the pilot. At each candidate N in $\{200, 250, 300, 350, 400, 500, 600, 700, 800\}$, N participants are resampled with replacement from the pilot (a participant drawn twice becomes two distinct participants), the outcome model of Section 4.1 is refitted per outcome, every confirmatory contrast is evaluated and the rule of Section 3 applied with Holm within a hypothesis; H4 is resampled the same way and the partner-choice model refitted. The rate of *supported* over 500 simulations per cell is the power estimate; no fit failed. Two effect sizes are simulated: the *pilot effect*, the resampled data as they are, and the *half effect*, every outcome shrunk toward its grand mean by half the pilot cell effect, which halves every contrast while keeping the noise and the participant structure (for H4, choices are regenerated from the pilot logistic fit with the common preference halved and each participant's deviation kept). The script is `src/r/power_analysis.R`; its decision-rate table is attached to the registration.

Decision rates at the pilot effect. Every confirmatory test is at 1.00 at every N on the grid (500 simulations per cell).

Test	200	250	300	350	400	500	600	700	800
H1, payoff	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H1, entropy	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H2, payoff	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H2, entropy	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H3, entropy	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H4, choice	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H5, payoff	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Decision rates at half the pilot effect:

Test	200	250	300	350	400	500	600	700	800
H1, payoff	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H1, entropy	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H2, payoff	.67	.72	.86	.91	.93	.96	.98	1.00	1.00
H2, entropy	.98	.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H3, entropy	.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H4, choice	.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H5, payoff	.95	.97	.99	1.00	1.00	1.00	1.00	1.00	1.00

Secondary claims. S1 to S3 are model comparisons and S4 a posterior quantity; none is powered, and each is reported with its paired standard error or posterior interval whatever N delivers.

Chosen sample size. We plan to recruit $N = 350$ participants who survive the exclusion rules of Section 6.4. At $N = 350$ the lowest-powered claim, H2 on payoff, is at 0.91 at half the pilot effect.

6.4 Exclusions

Every exclusion rule lives in one script, `src/r/preprocess_participant_data.R`, which writes a ledger of counts per rule (`exclusions.csv`). Nothing else is excluded. In particular no participant is dropped for poor performance, and the practice session plays no part in any rule.

	Level	Rule	Reason	Pilot E
X1	participant	returned the submission on Prolific	withdrawal	8 of 49
X2	participant	answered yes to Prolific's colour-blindness prescreen	game asteroids and rewards and partners are in various colours and otherwise may be hard to tell apart	2
X3	participant	completion in under 300 s	300 s is roughly the time it takes to click through the game as fast as possible; indicative of not paying attention	0
X4	participant	completion takes over 60 min	study left unattended	1
X5	session	no watch or play shot logged	no data	0
X6	session	less than 2 rounds or less than 4 games	incomplete session	0
X7	session	number of shots not exactly 6	incomplete or corrupted game	0
X8	session	page reloaded during game	a reload restarts the game giving them extra shots to learn from	0
X9	participant	more than one page load of the study	a reload restarts the study	0
X10	participant	no change to starting position	participant never moved	0
X11	participant	not exactly three sessions after exclusions	dropouts, missing or repeated sessions	0

A Prolific time-out is not an exclusion. The 60-minute window of X4 is measured on the game's own log, not by Prolific. Timed-out participants not otherwise

excluded are thus kept and paid.

7 Frozen artefacts, reporting and data availability

[TODO] [TODO]

Reporting. When the data are in, the results are written by the pipeline's report stage at the tagged commit into `results.md`, with no number typed by hand, in the fixed order of the pipeline's manifest (`numbers.py`): the sample and the exclusion ledger in the format of Section 6.4 against the registered $N = 350$; one row per claim of Section 3 with estimate, 95% interval and verdict, for S1 to S4 the check-gated final verdict of the report stage with the choice stage's own verdict beside it, and the reward rows beside their contrasts; the twelve-cell grid of means with intervals per outcome, solo's first and third games beside it; the ladder of every registered model with its paired difference from HOS, the Pareto-k counts and the sessions refitted under the fallback, and the per-participant differences; the gate diagnostics, recovery outcomes and predictive-check bands, whatever they show; each entry of Section 5 that was run, entries not run listed as such; and every deviation from this document with its reason, an empty list stated as such.